Bally

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Implement packet sniffing using Raw sockets

Aim: To implement packet sniffing using raw sockets in python and capture network packets

Procedure:

1. Open VS code and create a python file named packet sniffer.py
2. Import the required socket and struct libraries
3. Create and configure a raw socket using local IP address
4. Run VS code as administrator and execute the program.
5. Generate network traffic using ping 8.8.8.8.
6. Capture & display the source IP, destination IP, protocol and packet size

Result: Thus, packet sniffing was successfully using raw sockets in python and network packets were captured and their detail were displayed.

Program :

```
import socket
Sniffer = socket.socket(
    socket.AF_INET,
    socket.SOCK_RAW,
    socket.IPPROTO_IP
)
hostname = socket.gethostname()
local_ip = socket.gethostbyname(hostname)
print ("=" *45)
print ("RAW SOCKET PACKET SNIFFER")
print ("=" *45)
print ("Local IP:", local_ip)
print ("Waiting for packets ...")
print ("press Ctrl+C to stop.\n")
sniffer.bind ((local_ip, 0))
sniffer.setsockopt(
        socket.IPPROTO_IP,
        socket.IP_HDRINCL
sniffer.ioctl(socket.SIO_RCVALL,socket.RCVALL_ON)
try:
    while True:
        packet, address = sniffer.recvfrom(65535)
        print ("-" *45)
        print ("Source IP:", address[0])
        if len(packet) >=20:
            protocol = packet[9]
            if protocol ==6:
                print ("protocol : TCP")
            else:
                print ("protocol :", protocol)
finally:
    sniffer.ioctl(socket.SIO_RCVALL, socket.RCVALL_OFF)
    sniffer.close()
```